MatMek4270 Assignment 1

12. oktober 2026

1 1.2.3 Exact solution

The wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u \tag{1}$$

Substituting $u(t, x, y) = e^{i(k_x x + k_y y - \omega t)}$ into the wave equation gives:

$$\frac{\partial^2 u}{\partial t^2} = (-i\omega)^2 u = -\omega^2 u,$$

$$\frac{\partial^2 u}{\partial x^2} = (ik_x)^2 u = -k_x^2 u, \qquad \frac{\partial^2 u}{\partial y^2} = (ik_y)^2 u = -k_y^2 u$$

$$-\omega^2 u = -c^2(k_x^2 + k_y^2)u$$

As u(t, x, y) never will be zero, setting $k^2 = k_x^2 + k_y^2$:

$$\omega^2 = c^2 k^2 \tag{2}$$

Therefore u satisfies the Wave equation provided that $\omega^2 = c^2 k^2$

2 1.2.4. Dispersion coefficient

We have:

$$\frac{u_{i,j}^{n+1} - 2u_{i,j}^n + u_{i,j}^{n-1}}{\Delta t^2} = c^2 \left(\frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{h^2} \right)$$

Assuming $m_x = m_y$ such that $k_x = k_y$:

$$u_{i,j}^n = e^{\mathrm{i}(kh(i+j)-\tilde{\omega}\, n\Delta t)} \tag{3}$$

Inserting $u_{i,j}^n$ into the discretized equation, and also dividing both sides by $u_{i,j}^n$

$$\frac{e^{-\mathrm{i}\tilde{\omega}\Delta t} - 2 + e^{\mathrm{i}\tilde{\omega}\Delta t}}{\Delta t^2} = 2c^2 \frac{e^{\mathrm{i}kh} - 2 + e^{-\mathrm{i}kh}}{h^2} \tag{4}$$

Using: $e^{\mathrm{i}\theta} - 2 + e^{-\mathrm{i}\theta} = 2\cos\theta - 2 = -4\sin^2(\frac{\theta}{2})$

$$\frac{1}{\Delta t^2}\sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) = \frac{2c^2}{h^2}\sin^2\left(\frac{kh}{2}\right) \tag{5}$$

$$C = c\Delta t/h$$

Multiplying by Δt^2 and using $C = c\Delta t/h$:

$$\sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) = 2C^2\sin^2\left(\frac{kh}{2}\right) \tag{6}$$

Choosing $C = 1/\sqrt{2}$ gives $2C^2 = 1$, so

$$\sin^2\left(\frac{\tilde{\omega}\Delta t}{2}\right) = \sin^2\left(\frac{kh}{2}\right) \quad \rightarrow \quad \frac{\tilde{\omega}\Delta t}{2} = \frac{kh}{2} \tag{7}$$

$\tilde{\omega} = kh/\Delta t = ck/C = \sqrt{2}\,ck$. Which shows that for $C = 1/\sqrt{2}$ we get $\tilde{\omega} = \omega$